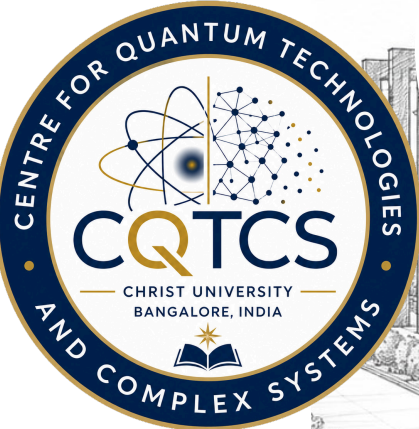
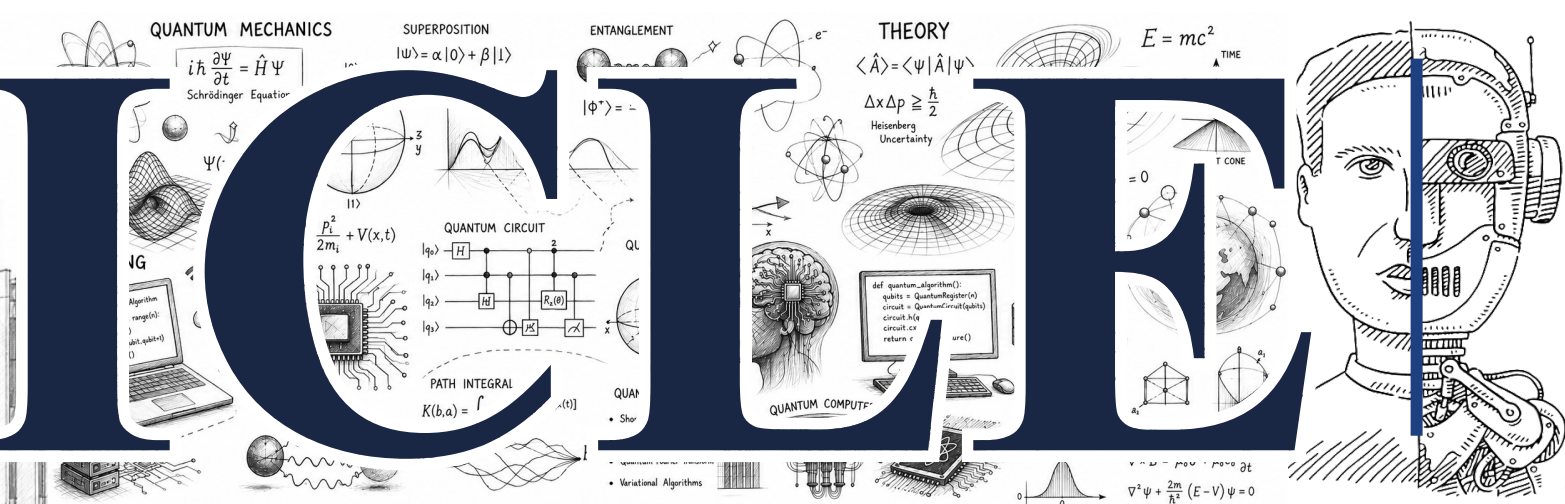


ISSUE 01



QUANICLE

THE NEWSLETTER



CHRIST
(DEEMED TO BE UNIVERSITY)
BANGALORE | DELHI NCR | PUNE

COORDINATOR'S NOTE

It is with great pleasure that we present the inaugural issue of QUANICLE, the newsletter of the Centre for Quantum Technologies and Complex Systems (CQTCS).

This platform will showcase our research excellence, celebrate achievements, and highlight opportunities for collaboration in the exciting world of quantum science and complex systems.

Together, let us continue to push the boundaries of knowledge and create a meaningful impact on society.

Dr. Mithun Thudiyangal
Coordinator, CQTCS



REPRESENTING THE CENTRE IN AUSTRIA



Ms. Dipasha Chokeda

12–15 July 2026



Mr. Sahil Satapathy

Innsbruck, Austria

Have been selected to participate in the Introductory Course on Ultracold Quantum Gases, jointly organized by Universität Innsbruck and the Institute for Quantum Optics and Quantum Information (IQOQI), Austria.

The program brings together young researchers from around the world to explore the fascinating physics of ultracold quantum gases.

EVENT RECAP

Dr. Sreeja C S attended the Quantum Pitch Fest, held on 22–23 May 2026 at the Indian Institute of Science (IISc), Bengaluru, organized by WIN-CoE, IISc Bengaluru, IQTI, QuRP, and the Foundation for QC Innovation.

MEET OUR GROUP LEADERS



Dr. Mithun Thudiyangal
Group Leader



Research Focus
Ultracold Quantum Matter,
Quantum Algorithms &
Quantum fluids



Dr. Basudeb Sain
Group Leader



Research Focus
Photonics / Quantum
Photonics & Optical
Metasurfaces



Dr. Arindam Mallick
Group Leader



Research Focus
Quantum Information,
Quantum Simulation &
Mathematical Physics



Dr. Sreeja C S
Group Leader



Research Focus
Quantum-Safe
Cryptography
& Cyber Security



Dr. Ranjith Kumar K M
Group Leader



Research Focus
Quantum Materials &
Quantum Magnetism

RESEARCH SPOTLIGHT

New Insights into Quantum Superfluid Dynamics

Dr. Mithun Thudiyangal and collaborators from the University of Luxembourg have published their research article, "[Kibble-Zurek Scaling and Spatial Statistics in Quenched Binary Bose Superfluids](#)," in [Communications Physics \(Nature\)](#),

highlighting significant advances in the study of non-equilibrium dynamics in quantum many-body systems.

The study explores universal features of far-from-equilibrium dynamics and introduces a framework for characterizing stochastic geometry in multicomponent quantum systems.

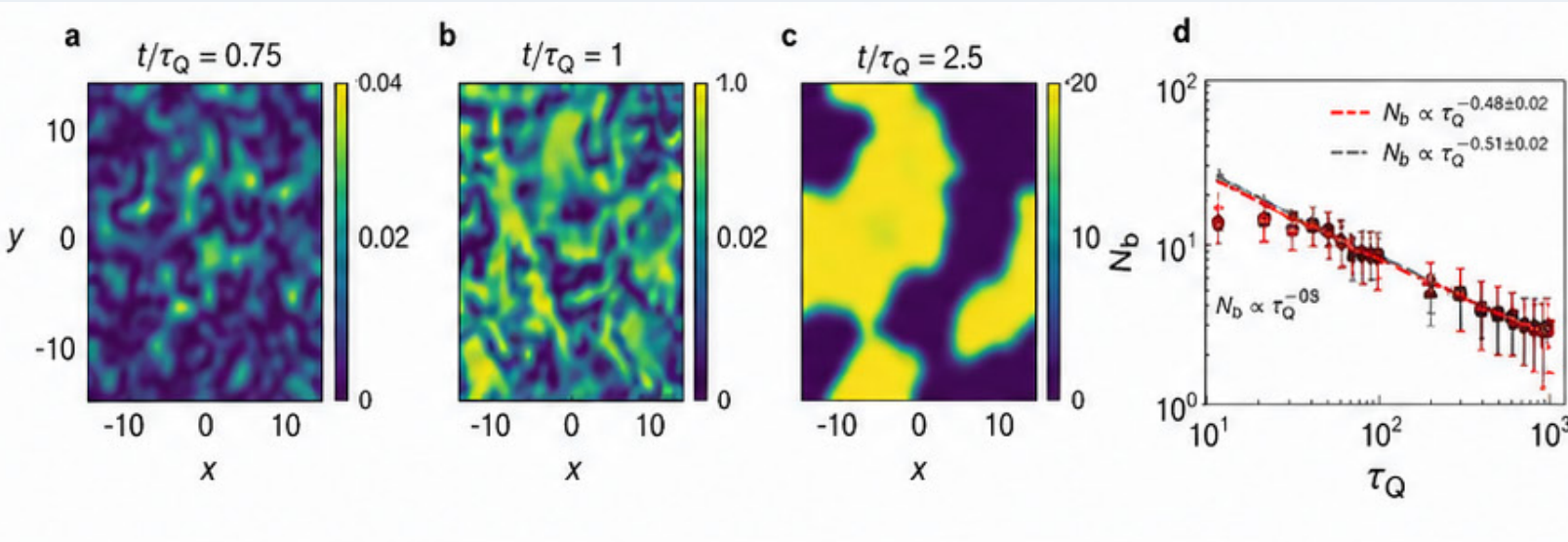
communications physics

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Kibble-Zurek scaling and spatial statistics in quenched binary Bose superfluids

Subhadeep Patra¹, Arko Roy¹, Seong-Ho Shinn², Adolfo del Campo^{2,3} & Mithun Thudiyangal⁴

KEY RESULTS



PUBLICATIONS



Dr. Sreeja C S

integrating contextual attributes, authentication strength, behavioral evidence, and retry dynamics. ZeTHAA utilizes a probabilistic risk model and dual-policy thresholds to partition outcomes into allow, step-up, and block regions, enabling precise control over security-usability trade-offs.

Provably Adaptive Trust Dynamics in Context-Aware Zero-Trust Systems: A Formal Framework for Continuous Verification

Publisher: IEEE

Vivin Krishnan, C. S. Sreeja, Volume: 14, Quartile: Q1, Impact Factor: 3.9

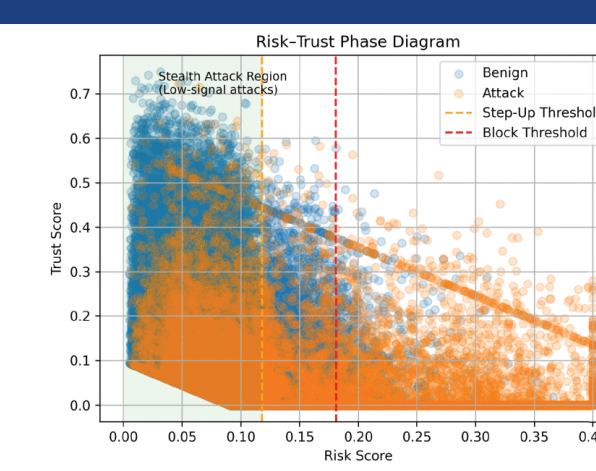
Cite This

Dr. Sreeja C. S. published a research article in the Q1 journal [IEEE Access](#), co-authored with Ph.D. scholar Mr. Vivin Krishnan, titled "[Provably Adaptive Trust Dynamics in Context-Aware Zero-Trust Systems: A Formal Framework for Continuous Verification](#)."

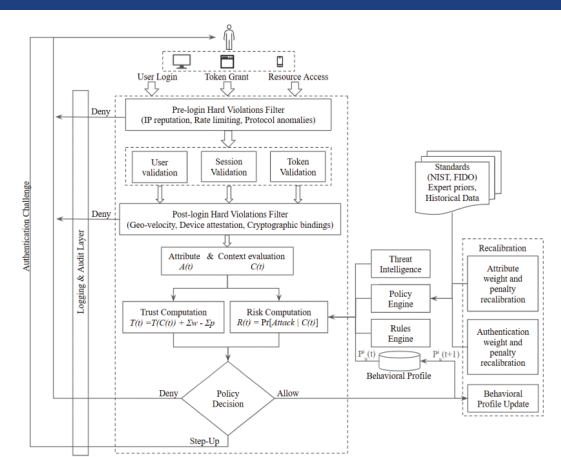
The paper introduces Zero-Trust Hybrid Adaptive Authentication (ZeTHAA), a continuous authentication and authorization framework

integrating contextual attributes, authentication strength, behavioral evidence, and retry dynamics. ZeTHAA utilizes a probabilistic risk model and dual-policy thresholds to partition outcomes into allow, step-up, and block regions, enabling precise control over security-usability trade-offs.

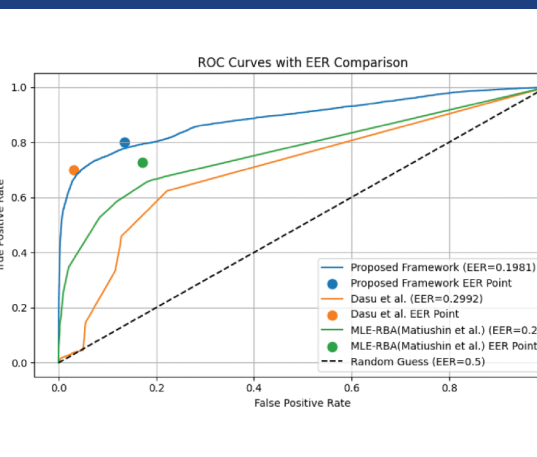
RISK-TRUST PHASE DIAGRAM



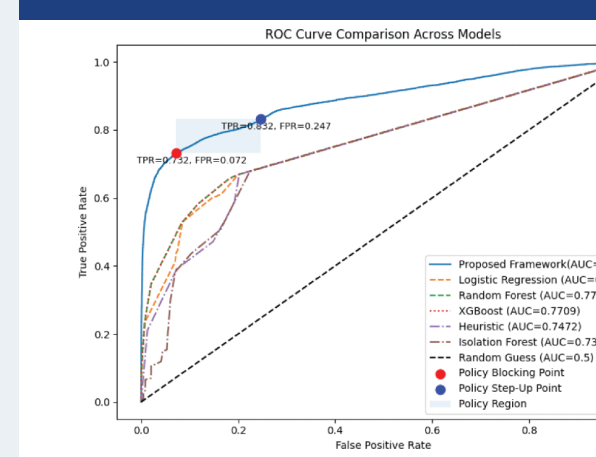
ARCHITECTURAL FRAMEWORK



ROC CURVE WITH EER POINTS



ROC CURVE COMPARISON



UPCOMING EVENTS



6 June | Certificate Course:
On Emerging Quantum Technologies
Online & Offline Modes

EDITORIAL TEAM

Designer & Editor
Abhijith E

Advisor
Dr. Sreeja C S

Committee Members

Kesiya John

Deepali Elizabeth